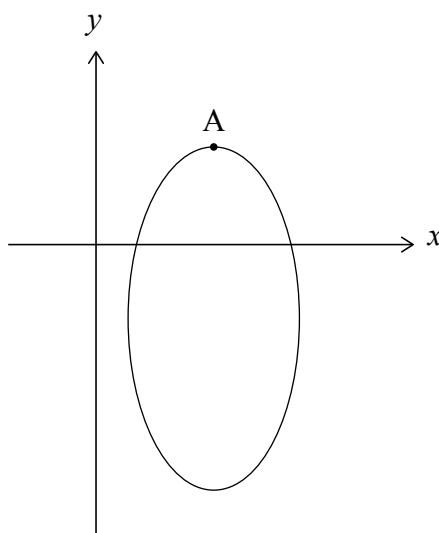


Do **not** write solutions on this page.

12. [Maximum mark: 22]

The curve C has equation $4x^2 + y^2 - 24x + 4y + 20 = 0$.

The following diagram shows C with a maximum point at A .



(a) Use implicit differentiation to show that $\frac{dy}{dx} = \frac{4(3-x)}{y+2}$. [4]

(b) Hence, determine the domain of C . Give your answer in the form $3 - \sqrt{a} \leq x \leq 3 + \sqrt{a}$, where $a \in \mathbb{Z}^+$. [4]

(c) Find (x_A, y_A) , the coordinates of A . [3]

A line $y = mx$ is a tangent to C , where $m \in \mathbb{Z}$.

(d) Find the possible values of m . [4]

The line $y = -4x$ touches C at point B .

(e) Find y_B , the y -coordinate of B . [3]

The region bounded by the curve C , the y -axis and the lines $y = y_A$ and $y = y_B$, is rotated 360° about the y -axis to form a solid of revolution.

(f) Find the volume of the solid formed. [4]

